

From Detection to Attribution: Physically-Coupled, Cross-Domain Monitoring of Ephemeral-Channel Geomorphic Change from Terrain Alone

NASA ROSES-2025 F.5: Future Investigators in NASA Earth and Space Science and Technology (FINESST)
Target division: Earth Science Division (EARTH25), Antarctic cryosphere / climate Proposal deadline: 14 July 2026 (11:59 PM EDT) Structure: Graduate student = Future Investigator (FI), primary author and intellectual lead; faculty advisor = PI of record. Award up to ~\$50,000/yr for up to 3 years.

Summary. This project will transform an existing Antarctic stream-change monitor into an explanatory and predictive tool. Building on Barlow (2026), which mapped two decades of ephemeral-channel geomorphic change across the McMurdo Dry Valleys, the proposed work will (O1) attribute that change to its physical/climate drivers, (O2) generalize the terrain-only detector across sensors and valleys, and (O3) deliver calibrated per-pixel uncertainty. The change-detection foundation is already operational and validated against Barlow's published uncertainty ranges, with an initial driver signal shown in §4.

1. Background and the Gap

The McMurdo Dry Valleys (MDVs) are Earth's largest ice-free Antarctic desert and, for a century, were considered the planet's most geomorphologically stable landscape. That assumption is breaking down: in an energy-limited system, small increases in solar absorption and permafrost degradation translate directly into ephemeral summer streams that flow for a few weeks each year, move sediment, and reshape channels. These streams are the most sensitive measurable indicator of climate change in the most stable place on Earth, a continental-scale "climate canary."

Barlow (2026) [1] established the measurement foundation in three transferable pillars:

1. Terrain-only semantic segmentation: a U-Net delineates stream boundaries from DEM derivatives alone (elevation, slope, aspect most informative; no water/color signal), so it works in channels that are dry 11 months a year. 2. Satellite-DEM validation: free REMA satellite DEMs are proven accurate enough for centimeter-to-meter stream-corridor change detection, using ICP (iterative closest point — sliding one 3-D surface onto another until they match) for alignment and NMAD (normalized median absolute deviation — a standard deviation that ignores outliers) for honest error bars; the errors follow a Laplacian (sharp-peaked, heavy-tailed) distribution rather than a Gaussian, which is why the robust statistic is required. 3. Multi-epoch change detection: subtract one epoch's DEM from another (DEM-of-Difference, DoD) inside the segmented stream masks, and count only elevation changes exceeding the level of detection ($\text{LOD}_{95} = 1.96 \times \text{NMAD}$ — the smallest change distinguishable from measurement noise at 95% confidence). This yields per-stream erosion/deposition and acceleration rates across four valleys and three epochs (2001 & 2014 lidar, 2021–23 REMA).

The gap. Barlow's dissertation ends at *description*: it maps where change is happening (Denton Hills is the erosion hotspot; parts of Taylor Valley are accelerating) but explicitly defers two threads to future work: (a) coupling geomorphic change to physical/climate drivers, and (b) generalizing the model beyond the MDVs. A Future Investigator project lives precisely in that gap: turning a *monitor* into an *explanatory and predictive* tool (Fig. 1). This is a genuine scientific increment, not a replication.

From detection (Barlow 2026) → attribution & prediction (this FINESST project)

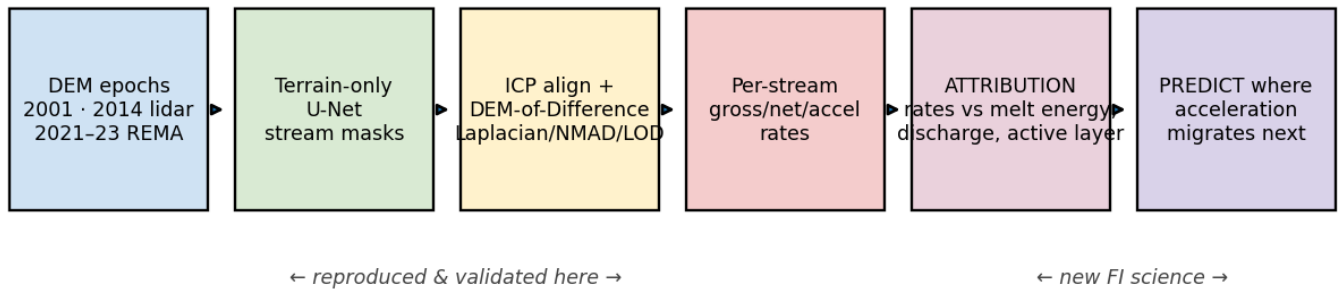


Figure 1. From detection (Barlow 2026; the first four stages) to the proposed attribution and prediction science (last two stages, the new contribution of this project).

2. Objectives and Hypotheses

#	Objective	Hypothesis	Increment over Barlow
O1	Attribution (headline science). Pair per-stream gross/net/acceleration rates with energy-balance drivers, air/ground temperature, insolation, positive-degree-days (PDD), melt-season discharge, glacier mass balance, active-layer depth, from MDV-LTER + ERA5/AMPS reanalysis.	H1: Geomorphic flux is governed by <i>cumulative melt energy</i> , not raw water volume; a PDD/insolation-based energy model predicts per-stream gross rate better than discharge alone (and explains the residual scatter in Fig. 4).	Listed verbatim as Barlow future work; never quantified.
O2	Cross-sensor & cross-valley generalization. Make the terrain-only segmenter robust across sensors (lidar 2001/2014 ↔ REMA) and all four valleys; quantify and correct the lidar→satellite domain shift that limits the satellite epoch.	H2: A sensor-aware normalization (per-pixel slope/aspect-conditioned bias correction) closes most of the lidar-REMA F1 gap, recovering near-lidar segmentation on REMA.	Barlow flags sensor generalization as future work.
O3	Calibrated uncertainty. Extend the Laplacian/NMAD/LOD framework into a per-pixel, sensor-aware confidence layer so every change map ships with calibrated uncertainty.	H3: Conditioning NMAD on slope, aspect, and sensor yields LOD maps whose empirical exceedance matches the nominal 95%, a trustworthy-EO product.	Barlow uses a single global NMAD per epoch-pair.

Predictive payoff (ties O1-O3 together): a model that, given a driver field, predicts *where acceleration migrates next*, the actionable form of the "climate canary."

3. Methodology

Data (free/public; see §6). Elevation backbone: MDV airborne lidar 2001 (NASA ATM, 2 m) + 2014 (NCALM, 1 m), and REMA v2.0 (2 m, 2021–23). Per-epoch derivatives, elevation, slope, aspect, profile curvature (signed), MFD flow accumulation, lidar intensity. Drivers: MDV-LTER meteorology (Lake Bonney), 21 stream-discharge gauges, glacier mass balance (7 glaciers), continuous soil T/EC/VWC (active-layer

proxy); ERA5 monthly 1993–2024 and AMPS 2.67-km for spatial gap-fill. Labels: MCM-LTER stream centerlines (public stand-in for Barlow's access-gated 217 hand-digitized tiles, see §7 risk).

O1, Attribution. For each gauged stream and epoch, the proposed work will (i) compute gross/net geomorphic rate by DoD inside the channel mask (operational, Fig. 2); (ii) build a per-stream energy time series, melt-season PDD and incoming shortwave from LTER + ERA5/AMPS, cumulative discharge, and active-layer depth; (iii) fit hierarchical regression / random-forest models of rate vs. drivers, with leave-one-stream-out cross-validation; and (iv) test H1 by comparing an energy-balance predictor against discharge-only (the raw-discharge relationship in Fig. 5 is real but incomplete). Acceleration (3-epoch Taylor streams) will be regressed against driver *trends*.

O2, Generalization. The work will quantify the lidar→REMA domain shift over stable terrain as a function of slope/aspect, learn a correction, retrain/fine-tune the segmenter with sensor augmentation, and evaluate F1 across all four valleys and both sensor regimes. The FI's prior work applying this architecture to a different landscape (§5) shows it transfers across sensors and biomes.

O3, Calibrated uncertainty. The work will replace the single global NMAD with NMAD conditioned on (slope, aspect, sensor), emit a per-pixel LOD raster, and validate calibration by checking that observed sub-LOD residual fractions match nominal confidence on held-out stable terrain (Fig. 4 shows the global version; Laplacian fits the residuals far better than Gaussian).

Reproducibility/QC. The pipeline is fully scripted, input data and all derivatives regenerate from source, with robust statistics and CRS checks enforced at every step (e.g., an early QC catch, a 2001 nodata bug inflating NMAD to 2.7 m, was identified and fixed).

4. Preliminary Results

The change-detection foundation the proposed work builds on is already operational. A reproduction of Barlow's change detection on Taylor Valley falls inside her published uncertainty ranges (below), and an initial driver analysis surfaces the first attribution signal that O1 will develop.

(a) Change detection across all three epochs (Fig. 2).

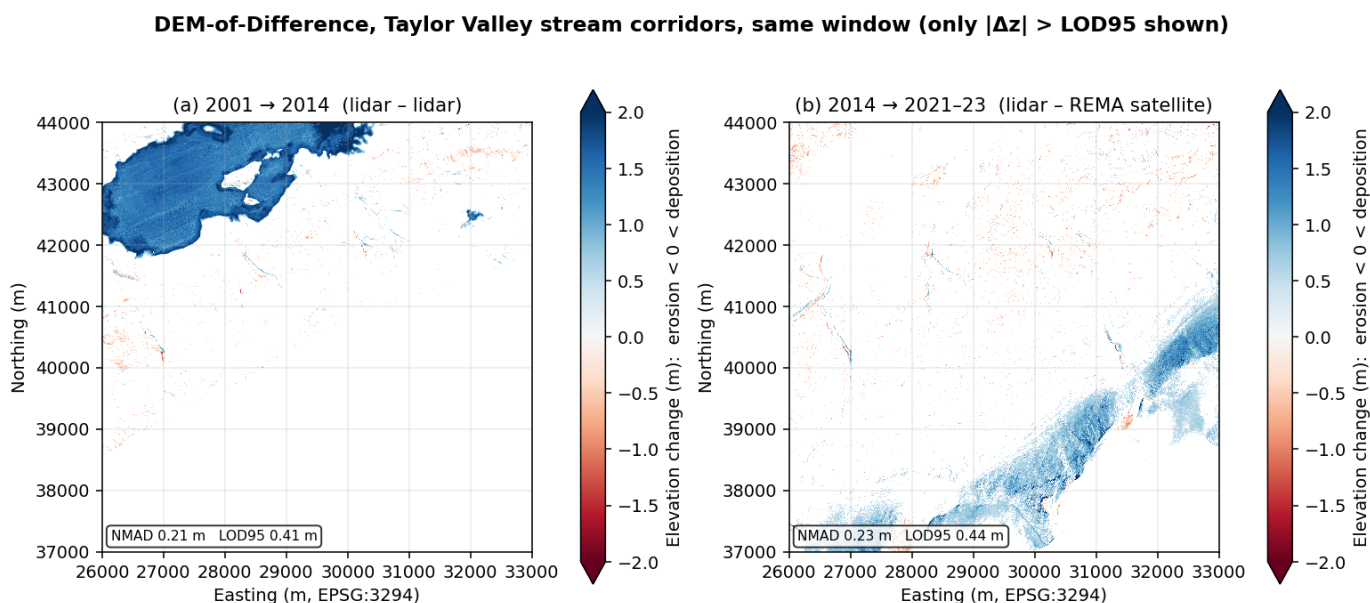


Figure 2. Elevation change over the same Taylor Valley stream-corridor window in two periods: (a) 2001→2014, comparing two lidar surveys; (b) 2014→2021-23, comparing lidar against the REMA satellite DEM. Red = surface lowered (erosion), blue = surface raised (deposition). Pixels are drawn only where the change exceeds the detection floor (LOD95) — i.e., is too large to be measurement noise — and coherent erosion/deposition patterns emerge above that floor. The broad flat patch in (a) is not sediment at all:

Lake Fryxell rose ~1.5 m between surveys, and an automated standing-water screen detects such flat water surfaces and excludes them from every stream-rate statistic.

Epoch pair	NMAD	LOD95	Barlow's published range	In range?
2001→2014 (lidar-lidar)	0.21 m	0.41 m	NMAD 0.07-0.46 / LOD95 0.15-0.92	✓
2014→2021-23 (lidar-REMA)	0.23 m	0.44 m	NMAD 0.19-0.53 / LOD95 0.37-1.04	✓

ICP alignment behaved exactly as Barlow's published version does: it tightened the error budget on steep, high-relief windows and correctly added no benefit on the flat valley floor (alignment needs terrain shape to grip onto). Convergence fitness was 0.95 m² mean-squared point distance.

(b) Per-stream rates (Fig. 3), the masked products O1 consumes.

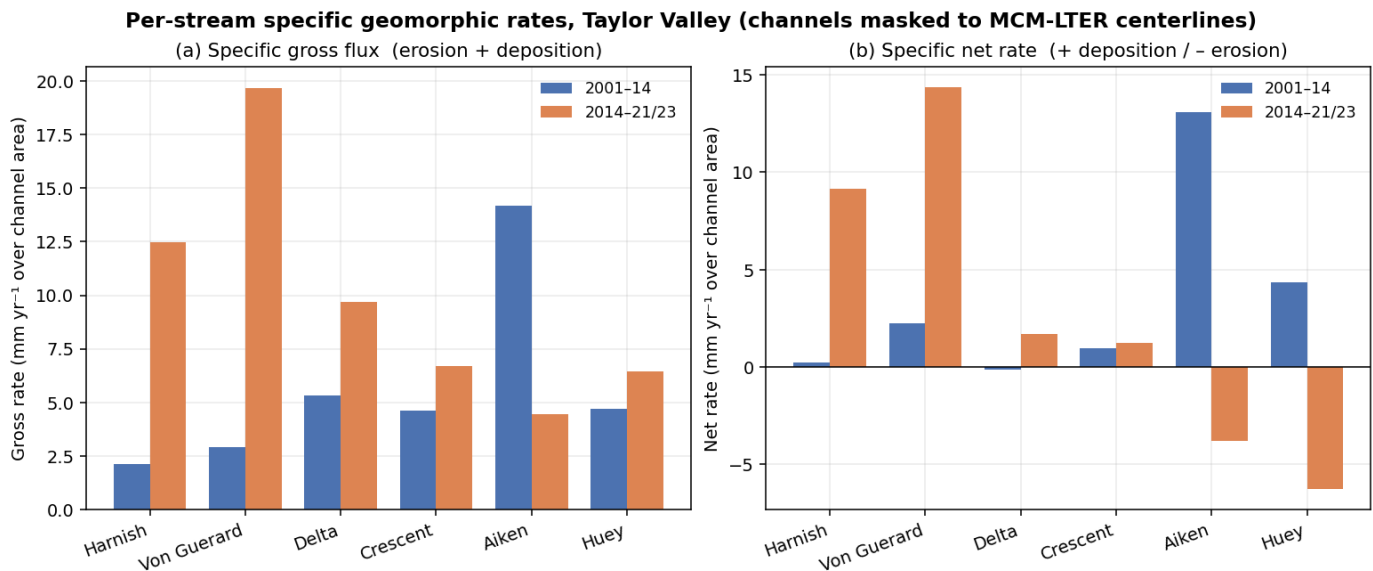


Figure 3. How much sediment each of six gauged Taylor Valley streams moved, per year, in each period. Rates are reported per square meter of channel (mm/yr of surface change) rather than as raw m³/yr totals: the surveys do not all cover the same footprint (the 2001 lidar swath is narrower than the satellite coverage), and a per-area rate cannot be inflated simply because one survey saw more ground.

(c) Robust error model (Fig. 4), the basis for O3.

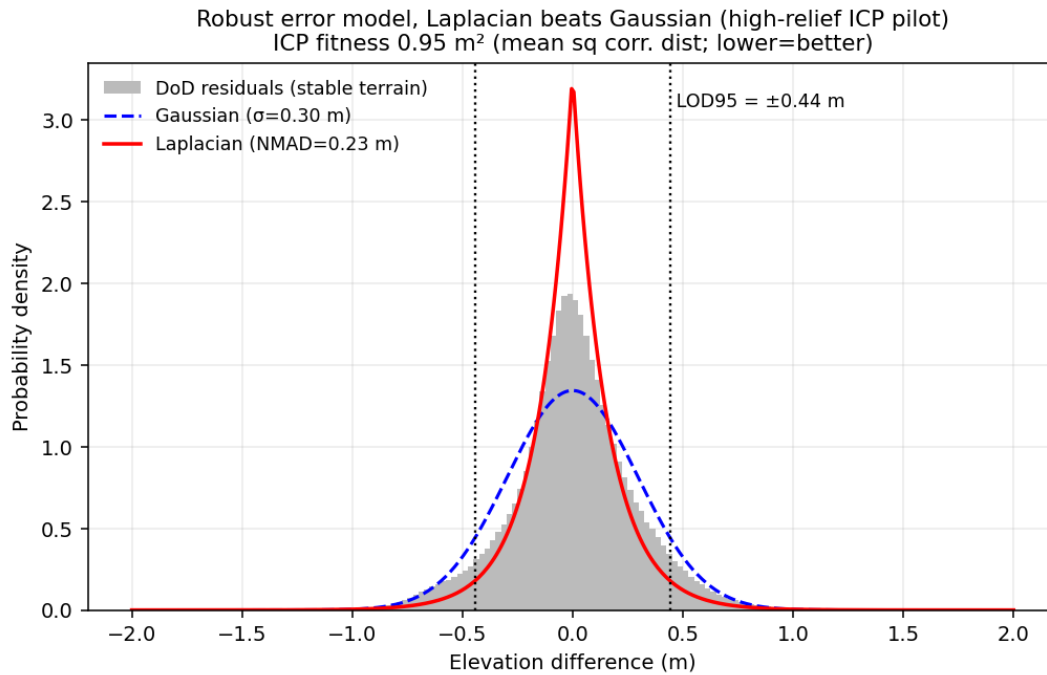


Figure 4. The measurement noise, profiled on ground that should not have changed at all. The error histogram is sharply peaked with heavy tails (a Laplacian shape), not the familiar bell curve: a handful of large outliers would wildly inflate a naive standard-deviation-based detection threshold, which is why the outlier-resistant NMAD is used instead. This reproduces Barlow's error-model finding and is the launch point for the proposed per-pixel calibrated-uncertainty layer (O3).

(d) A first attribution signal (Fig. 5).

Preliminary attribution: sediment flux vs melt discharge, n = 6 gauged streams per epoch (lake-level rise screened)

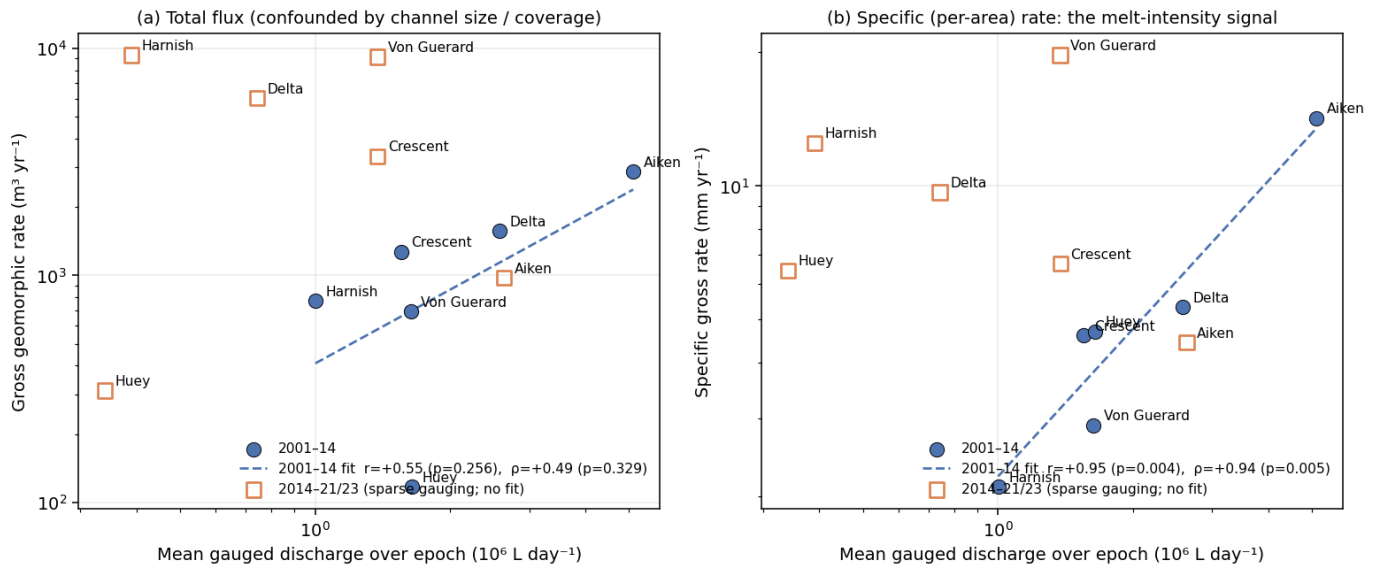


Figure 5. The first attribution signal: each stream's sediment-movement rate plotted against how much meltwater it carried (mean gauged discharge; both axes logarithmic; the lake signal from Fig. 2 screened out). In the 2001→2014 period the relationship is strong — correlation $r = +0.95$ ($p = 0.004$), rank correlation $\rho = +0.94$ ($p = 0.005$) — and it survives dropping any single stream, so it is not driven by one outlier. The satellite period shows no coherent relation, but for an instructive reason: after 2015 the stream gauges ran only sporadically (48–362 recorded days per stream), so the water axis itself is unreliable there, and no trend line is fit. That is precisely the motivation for O1: where gauge data are dense the melt-sediment link is strong, so the proposed work replaces patchy gauging with modeled melt

energy (degree-days, sunlight, thaw depth) to extend the analysis across all epochs and valleys.

5. FI Qualifications

The FI has independently built and operated this exact toolchain, U-Net semantic segmentation on lidar-derived terrain rasters, ICP co-registration, and DEM-of-Difference change analysis, on an unrelated landscape (channels, roads, and disturbance scars detected from elevation alone in the Appalachian Plateau). That prior work shows the architecture transfers across sensors and biomes, directly de-risking O2, and that the FI already commands the full pipeline end to end.

6. Data Requirements and Availability

Every dataset required for the proposed work is free/public and already in hand (regenerable from source via scripted pipelines), so data availability poses no schedule risk. The single access-gated dependency is Barlow's training labels (mitigated below and in §7).

Dataset (source)	What it shows	Role in the proposed work	Status
2001 airborne lidar, 2 m (NASA ATM, via USGS)	Valley-floor surface in 2001	Baseline epoch for all change detection	✓ in hand
2014 airborne lidar, 1 m (NCALM, via OpenTopography)	Same terrain 13 yr later at benchmark accuracy	Reference epoch; anchors the error model	✓ in hand
REMA satellite DEM, 2 m, 2021-23 (PGC/Maxar stereo)	Most recent surface, continent-wide	Extends the record past the last lidar flight; the sensor O2 must generalize to	✓ in hand
Per-epoch terrain derivatives (slope/aspect/curvature/MFD-flow/intensity)	Channel-diagnostic terrain form	Input features the U-Net segmenter reads	✓ scripted; demonstrated (Taylor pilot)
LTER stream gauges, 21 streams, daily, 1990s-present (EDI)	Meltwater each stream carried	Direct discharge driver for O1 (source of the Fig. 5 signal)	✓ in hand
LTER meteorology (EDI)	Air temperature, radiation, wind	PDD + insolation melt-energy drivers (O1/H1)	✓ in hand
LTER glacier mass balance, 7 glaciers (EDI)	Seasonal ice gain/loss per glacier	Links stream water supply to its source	✓ in hand
LTER soil temperature/moisture (EDI)	Summer thaw depth	Active-layer driver (O1/H1)	✓ in hand
ERA5 monthly 1993-2024 (Copernicus CDS); AMPS 2.67 km (NCAR GDEX)	Continuous modeled atmosphere	Gap-fills drivers between stations and to ungauged streams; AMPS cross-checks ERA5	✓ in hand
MCM-LTER stream channel polygons (EDI knb-lter-mcm.6007)	Mapped channel of every named stream	Per-stream rate masks (used in Figs. 3/5) + label stand-in	✓ in hand
Barlow's 217 hand-digitized tiles (author-gated)	Expert-traced channel boundaries	Gold-standard U-Net training labels	▲ lone access-gated dependency (see §7)
Cross-valley DEMs, all four valleys (OpenTopography/PGC)	Terrain beyond Taylor Valley	O2 generalization test bed	✓ DEMs in hand; per-valley stacks to build
WorldView/Maxar imagery; published rates (PGC restricted; literature)	Visual ground truth; independent rates	Optional label QC / cross-check vs Barlow	optional

7. Management, Timeline, Risks, and Data Management Plan

FI role & development. The FI is intellectual lead and primary author; the advisor (PI of record) provides domain mentorship. Development plan: present at AGU (cryosphere) and an ISAES/SCAR venue; co-author the attribution paper (O1) as a stand-alone publication; complete graduate coursework in Bayesian/hierarchical modeling and remote-sensing uncertainty.

Timeline (3 years).

Period	Milestones
Yr 1	Full driver harmonization (PDD/insolation/active-layer series per gauged stream); O1 attribution model on Taylor Valley (extending the Fig. 5 signal); calibrated-uncertainty prototype (O3). Deliverable: attribution manuscript drafted.
Yr 2	Cross-sensor domain-shift correction + segmenter generalization across all four valleys (O2); apply attribution model basin-wide; AGU presentation. Deliverable: O2 paper.
Yr 3	Predictive model (where acceleration migrates next); 3-epoch acceleration attribution; calibrated-uncertainty product release (O3). Deliverable: synthesis/prediction paper + public data products.

Risks & mitigations. (1) *Barlow's label tiles are access-gated* → mitigation: LTER centerlines already work as a QC stand-in (Fig. 2–3 built on them); plan to request the tiles and, failing that, re-digitize a comparable training set (the FI's prior work shows label generation is in hand). (2) *REMA's uneven post-2014 coverage* → prioritize Taylor Valley (full 3-epoch coverage) for acceleration; treat other valleys as 2-epoch. (3) *Discharge gauge gaps* → ERA5/AMPS energy reanalysis fills spatial/temporal holes (a core reason O1 uses melt energy, not just gauges).

Data Management Plan. All input data are free/public (NASA ATM, NCALM/OpenTopography, PGC REMA, MCM-LTER/EDI, Copernicus ERA5, AMPS/GDEX), cited per source. Derived products, change-detection rasters, per-stream rate tables, calibrated-uncertainty layers, and stream-boundary polygons, will be released open-access (Dols via Zenodo / EDI) with the processing code (PDAL/WhiteboxTools/GDAL + Python). Heavy regenerable rasters are excluded from version control; lightweight scripts, figures, and tables are tracked. Outputs follow ASPRS/OGC standards and carry explicit CRS, method, and uncertainty metadata (per NASA open-science/trustworthy-EO guidance).

References

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Draft S/T/M section. Figures generated by barlow/build/_finesst_figures.py. Not a submitted proposal.